

Unit-4

Lecture-1

Schrodinger's Time Independent Wave Equation (STIE)

$$\frac{\partial^2 \Psi}{\partial x^2} + \frac{2m}{\hbar^2} (E - V) \Psi = 0$$

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V \Psi = E \Psi$$

- This equation is independent of time and gives a steady value.
- It is particularly useful when the energy of the particle is very small as compared to its rest energy.
- In most atomic problems, energy of the particle is very small when compared to rest energy.

Schrodinger's Time Dependent Wave Equation (STDE)

$$i\hbar \frac{\partial \Psi}{\partial t} = \frac{-\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V \Psi$$

- The solution of time dependent Schrodinger equation results in states which are not stationary.
- The corresponding probability densities are not constant and they vary with time.
- Let free electron in a box. If we apply an electric field on the box periodically, the original wave functions and the corresponding probability densities of the electron are not constant and they do depend on the time.

STIE

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V \Psi = E \Psi$$

The solution of time independent Schrodinger equation results in stationary states, where the probability density is independent of time.

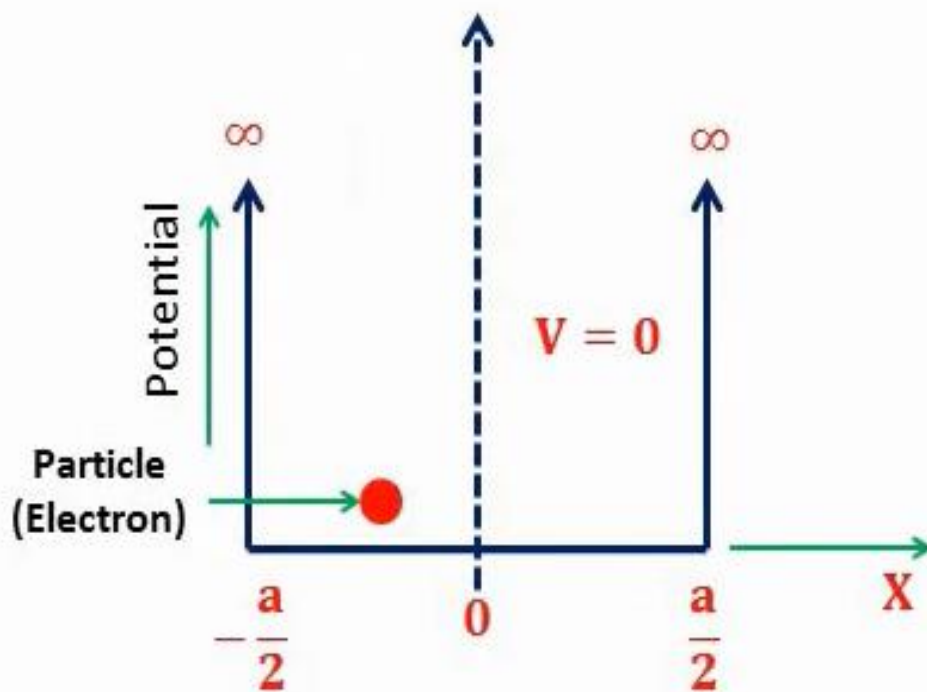
STDE

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V \Psi$$

The solution of time dependent Schrodinger equation results in states which are not stationary. The corresponding probability densities vary with time.

As Newton's laws predict the future behavior of a dynamic system in classical mechanics, Schrodinger's equations are used to predict future behavior in quantum mechanics.

Particle (Electron) in an infinite potential Well

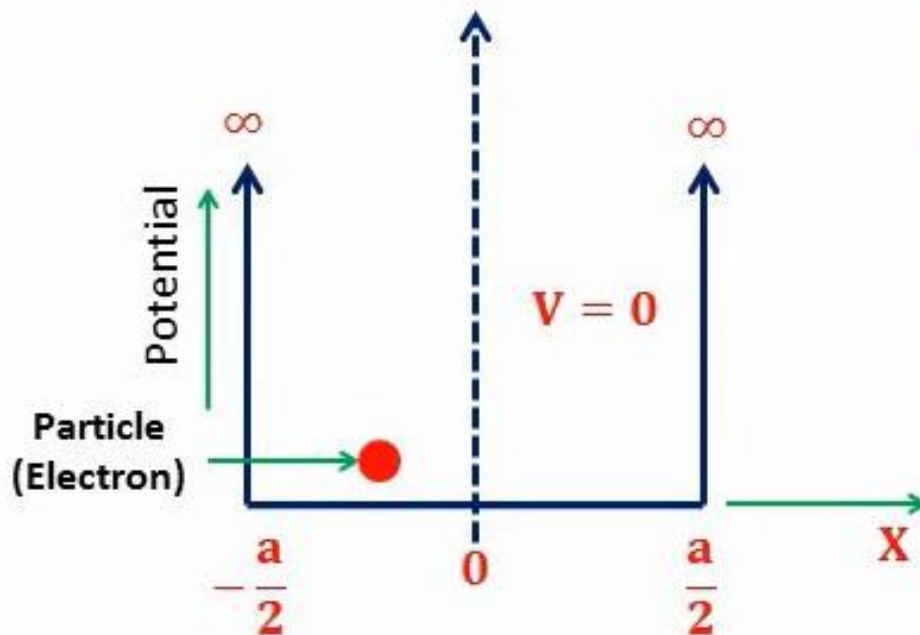


The particle is free to move within the walls from $-\frac{a}{2}$ to $\frac{a}{2}$ as potential in that region is **zero**.

At boundaries, potential is ∞ . So, the particle can neither cross the boundaries nor it can be located at $x = \pm \frac{a}{2}$

The wave function Ψ and its derivatives associated with the particle are finite, continuous and single valued within $-\frac{a}{2} < x < \frac{a}{2}$ but discontinuous at boundaries and does not exist outside the boundaries.

Particle (Electron) in an infinite potential Well



Schrodinger's Time Independent Equation is –

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi(x)}{\partial x^2} + V(x) \Psi(x) = E \Psi(x)$$

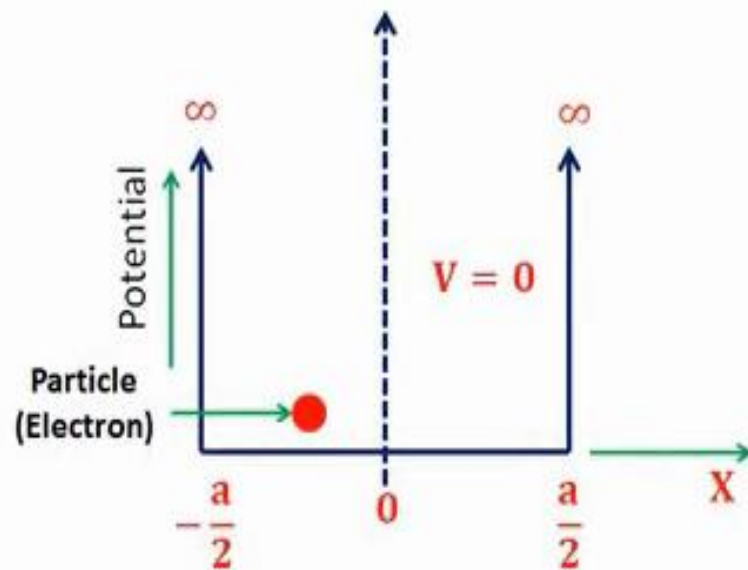
For region between $-\frac{a}{2} < x < \frac{a}{2}$, $V = 0$

$$\therefore -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi(x)}{\partial x^2} = E \Psi(x)$$

$$\therefore \frac{\partial^2 \Psi(x)}{\partial x^2} + \frac{2mE}{\hbar^2} \Psi(x) = 0$$

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Particle (Electron) in an infinite potential Box



$$\therefore \frac{\partial^2 \Psi(x)}{\partial x^2} + \frac{2mE}{\hbar^2} \Psi(x) = 0$$

1

Real solution of equation 1 is -

$$\begin{aligned} \Psi(x) &= A \sin kx + B \cos kx & \text{for } -\frac{a}{2} < x < \frac{a}{2} \\ \Psi(x) &= 0 & \text{for } |x| \geq \frac{a}{2} \end{aligned}$$

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At the boundaries, $\Psi(x) = 0$

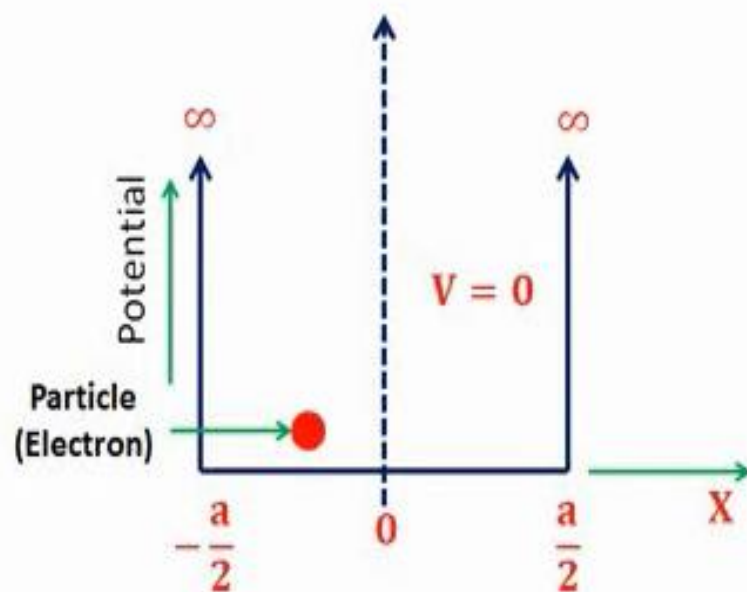
$$\therefore \Psi\left(x = -\frac{a}{2}\right) = -A \sin \frac{ka}{2} + B \cos \frac{ka}{2} = 0$$

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$$\therefore \Psi\left(x = \frac{a}{2}\right) = A \sin \frac{ka}{2} + B \cos \frac{ka}{2} = 0$$

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Particle (Electron) in an infinite potential Well



$$\therefore \frac{\partial^2 \Psi(x)}{\partial x^2} + \frac{2mE}{\hbar^2} \Psi(x) = 0 \quad (1)$$

$$\therefore \Psi\left(x = -\frac{a}{2}\right) = -A \sin \frac{ka}{2} + B \cos \frac{ka}{2} = 0 \quad (3)$$

$$\therefore \Psi\left(x = \frac{a}{2}\right) = A \sin \frac{ka}{2} + B \cos \frac{ka}{2} = 0 \quad (4)$$

$$(3) + (4)$$

$$2B \cos \frac{ka}{2} = 0$$

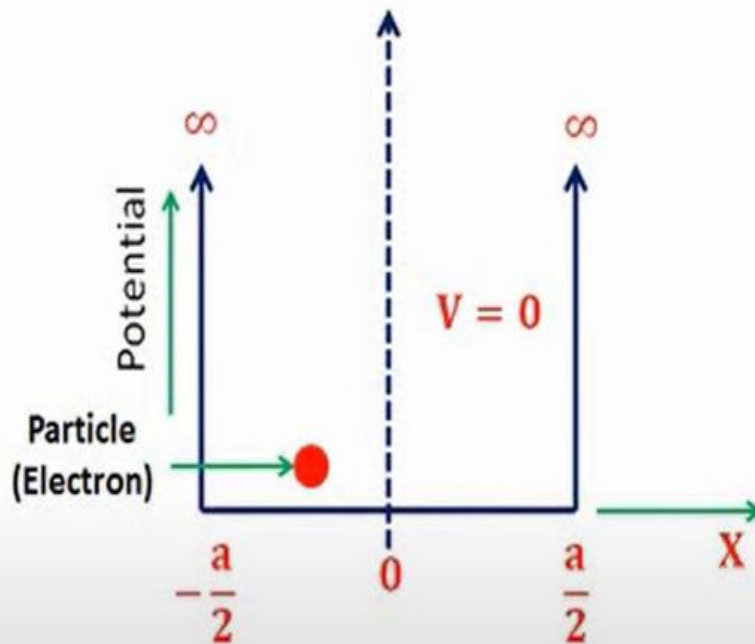
$$\therefore k = \frac{n\pi}{a} \\ n = 1, 3, 5, \dots \quad (5)$$

$$(3) - (4)$$

$$2A \sin \frac{ka}{2} = 0$$

$$\therefore k = \frac{n\pi}{a} \\ n = 2, 4, 6, \dots \quad (6)$$

Particle (Electron) in an infinite potential Well



$$\therefore k = \frac{n\pi}{a}$$

$$n = 1, 3, 5, \dots$$

5

$$\therefore k = \frac{n\pi}{a}$$

$$n = 2, 4, 6, \dots$$

6

$$\therefore k = \frac{n\pi}{a} ; n = 1, 2, 3, 4, \dots$$

Momentum of the particle is given by -

$$p_n = \hbar k = \hbar \frac{n\pi}{a}$$

Energy of the particle is given by -

$$E_n = \frac{(p_n)^2}{2m} = \frac{n^2 \hbar^2 \pi^2}{2ma^2} = \frac{n^2 h^2}{8ma^2}$$

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Thus, energy of particle (electron) trapped in infinite potential box (well) is quantized.

An electron is bound in a one dimensional potential well of width $2 \text{ \AA}$ and of infinite height. Find its energy values in ground state and first two excited states.

Given : $m = 9.1 \times 10^{-31} \text{ kg}$,

$$a = 2 \times 10^{-10} \text{ m}$$

$$h = 6.63 \times 10^{-34} \text{ J-s},$$

$$E_0, E_1, E_2 = ?$$

$$\text{We know, } E = \frac{n^2 h^2}{8 m a^2}$$

For ground state, $n = 1, E = E_0$

$$\begin{aligned} \therefore E_0 &= \frac{n^2 h^2}{8 m a^2} \\ &= \frac{1^2 \times (6.63 \times 10^{-34})^2}{8 \times 9.1 \times 10^{-31} \times (2 \times 10^{-10})^2} \\ &= 1.5 \times 10^{-18} \text{ J} \end{aligned}$$

For first excited state, $n = 2, E = E_1$

$$\begin{aligned} \therefore E_1 &= \frac{n^2 h^2}{8 m a^2} \\ &= \frac{2^2 \times (6.63 \times 10^{-34})^2}{8 \times 9.1 \times 10^{-31} \times (2 \times 10^{-10})^2} \\ &= 6 \times 10^{-18} \text{ J} \end{aligned}$$

For second excited state, $n = 3, E = E_2$

$$\begin{aligned} \therefore E_2 &= \frac{n^2 h^2}{8 m a^2} \\ &= \frac{3^2 \times (6.63 \times 10^{-34})^2}{8 \times 9.1 \times 10^{-31} \times (2 \times 10^{-10})^2} \\ &= 1.35 \times 10^{-17} \text{ J} \end{aligned}$$

An electron is trapped in a one dimensional box of length 0.1 nm.
Calculate the energy required to excite the electron from its ground state to the 4th excited state.

Given : $m = 9.1 \times 10^{-31} \text{ kg}$,
 $a = 0.1 \times 10^{-9} \text{ m}$,
 $h = 6.63 \times 10^{-34} \text{ J-s}$,

$$E_4 - E_0 = ?$$

$$E = \frac{n^2 h^2}{8 m a^2}$$

For ground state, $n = 1, E = E_0$

$$\begin{aligned} \therefore E_0 &= \frac{n^2 h^2}{8 m a^2} \\ &= \frac{1^2 \times (6.63 \times 10^{-34})^2}{8 \times 9.1 \times 10^{-31} \times (0.1 \times 10^{-9})^2} \\ &= 6.04 \times 10^{-18} \text{ J} \end{aligned}$$

For fourth excited state, $n = 5, E = E_4$

$$\begin{aligned} \therefore E_4 &= \frac{n^2 h^2}{8 m a^2} \\ &= \frac{5^2 \times (6.63 \times 10^{-34})^2}{8 \times 9.1 \times 10^{-31} \times (0.1 \times 10^{-9})^2} \\ &= 1.51 \times 10^{-16} \text{ J} \end{aligned}$$

$$\begin{aligned} E_4 - E_0 &= 1.51 \times 10^{-16} - 6.04 \times 10^{-18} \\ &= 1.45 \times 10^{-16} \text{ J} \\ &= \frac{1.45 \times 10^{-16}}{1.6 \times 10^{-19}} = 906.25 \text{ eV} \end{aligned}$$

An electron is bound by a potential which closely approaches an infinite square well of width 2.5×10^{-10} m. Calculate the first lowest permissible energy for the electron.

Given : $m = 9.1 \times 10^{-31}$ kg,

$$a = 2.5 \times 10^{-10} \text{ m}$$

$$h = 6.63 \times 10^{-34} \text{ J-s}$$

For lowest permissible Energy level, $E = E_1$ and $n = 2$

$$E = \frac{n^2 h^2}{8 m a^2}$$

For lowest permissible Energy level, $n = 2$, $E = E_1$

$$\begin{aligned} \therefore E_1 &= \frac{n^2 h^2}{8 m a^2} = \frac{2^2 \times (6.63 \times 10^{-34})^2}{8 \times 9.1 \times 10^{-31} \times (2.5 \times 10^{-10})^2} = 3.864 \times 10^{-18} \text{ J} \\ &= 24.15 \text{ eV} \end{aligned}$$